

University of Leeds
School of Electronic and Electrical Engineering

ELEC5870M Interim Report

**Fabrication of Enhancement Mode HEMT Devices via Gate
Recession**

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Abstract

Abstract goes here

Acknowledgements

Abbreviations

2DEG	Two-dimensional electron gas
HEMT	High-electron-mobility transistor

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CHAPTER 1

Introduction

Modern communication, sensing, and microwave systems impose increasingly demanding requirements on the semiconductor devices that form their front-end signal paths. Higher operating frequencies, tighter noise constraints, and strict linearity targets continue to push transistor technologies beyond the capabilities of many silicon-based platforms. [1] Although silicon processes have made notable progress in recent years, they often fall short in providing the necessary electron mobility, noise performance, or high-frequency gain required for performance-critical microwave and millimetre-wave applications. (e.g. silicon's $1200 \text{ cm}^2/\text{Vs}$ compared to GaAs's $6500 \text{ cm}^2/\text{Vs}$ in transistors for MMICs) [1]

III–V High–Electron–Mobility Transistors (HEMTs) play a central role in addressing these challenges. Their heterostructure–based design creates a highly conductive channel at the interface between two semiconductor layers with different bandgaps. At this interface a two–dimensional electron gas (2DEG) forms, supporting exceptionally high carrier mobility. The resulting strong transconductance, stable high–frequency behaviour and low–noise performance have firmly established HEMTs as core devices throughout microwave and millimetre–wave circuit design. [2]

Research activity within the field has increasingly shifted toward gallium nitride (GaN) HEMTs, driven by the material properties enabled by its wide bandgap. GaN supports high breakdown voltages, strong power–handling capability and robust operation under large electric fields, making it the preferred choice for many high–power and high–linearity applications [3]. Despite this growing dominance, gallium arsenide (GaAs) HEMTs remain one of the most mature and well characterised III–V transistor platforms. Their stability, extensive documentation and long industrial history continue to make them an invaluable system for studying the relationships between heterostructure design, device architecture and electrical behaviour. [4, 5]

For the purposes of this project, GaAs presents clear practical and technical advantages. The Leeds Nanotechnology Cleanroom maintains established, reliable processes for GaAs/AlGaAs heterostructures, and suitable wafers are readily available with reproducible epitaxial quality [6].

Standard GaAs/AlGaAs HEMTs are inherently depletion-mode devices (d-mode); they naturally conduct current at zero gate bias and require a negative voltage to switch off. [2] This 'normally-on' behaviour is often undesirable as it necessitates complex negative polarity power supplies [7] and complicates the design of direct-coupled logic circuits. [8] Consequently, there is significant interest in developing enhancement-mode devices (e-mode) that are 'normally-off' and offer reduced power consumption.

One common approach to achieving e-mode operation is to recess the gate into the AlGaAs donor layer, thereby reducing the amount of charge carriers available to form the channel and shifting the threshold voltage in the positive direction. [9] However, achieving e-mode operation through gate recessing is inherently non-trivial, as it requires precise and repeatable removal of only a few nanometres of material, with small variations in etch depth leading to significant changes in threshold voltage and device behaviour. [10]

The primary objective of this project is to investigate the relationship between gate recess depth and threshold voltage in GaAs HEMTs, with a focus on understanding the practical limitations and repeatability of the fabrication process. This will be achieved through the development of a gate recession process in which the gate contact is recessed into the AlGaAs donor layer to shift the threshold voltage into the positive region.

To provide a rigorous comparison between d-mode and e-mode devices, this interim report first focuses on the fabrication and characterisation of a series of non-recessed d-mode devices, achieved in first semester. (Chapters 3 and 4). These standard d-mode devices serve as a control to validate the HEMT fabrication process and establish reference parameters throughout this project. Finally, Chapter 5 outlines the theory and proposed process strategy required to implement the precise donor layer etch needed to achieve e-mode operation in the second phase of this work.

CHAPTER 2

Technical Background

2.1 Overview of GaAs/AlGaAs HEMTs

In GaAs/AlGaAs high-electron-mobility transistors (HEMTs), the interface between the GaAs channel layer and the overlying AlGaAs barrier produces a discontinuity in the conduction band due to the differing bandgaps of the two materials. This offset creates a potential well on the GaAs side of the junction, as illustrated in Figure 2.1, which confines electrons to the heterointerface and enables the formation of a two-dimensional electron gas (2DEG).

Electrons supplied by silicon donors in the modulation-doped AlGaAs layer transfer into this potential well and accumulate to form the 2DEG, which serves as the conducting channel of the device. The spatial separation of carriers from ionised dopants results in exceptionally high electron mobility, underpinning the strong transconductance, low-noise performance, and high-frequency capability that distinguish HEMTs from bulk-channel field-effect transistors [11].

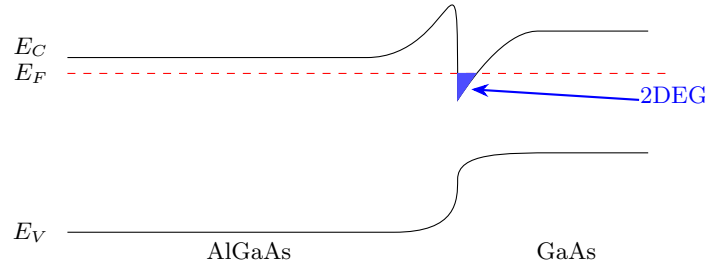


Figure 2.1: Band Structure Diagram of the AlGaAs/GaAs interface demonstrating the resulting Two-Dimensional Electron Gas (2DEG) formed in a Potential Well.

2.2 The GaAs/AlGaAs Heterostructure

The devices investigated in this project employ a conventional GaAs/AlGaAs heterostructure in which a sequence of epitaxial layers is engineered to create and control a 2DEG. A schematic cross-section of the structure is shown in Figure 2.2, highlighting the cap, donor, spacer, and channel layers whose thicknesses and compositions govern the electrostatic control and transport behaviour of the device.

At the surface of the structure lies a 10 nm thick, highly doped GaAs cap layer. This cap provides a chemically stable surface for forming the Schottky gate contact whilst protecting the underlying AlGaAs. Its high conductivity reduces series resistance in the access regions adjacent to the source and drain contacts, improving carrier injection into the channel.

Beneath the cap layer is a 40 nm thick silicon-doped AlGaAs donor layer. Silicon donors

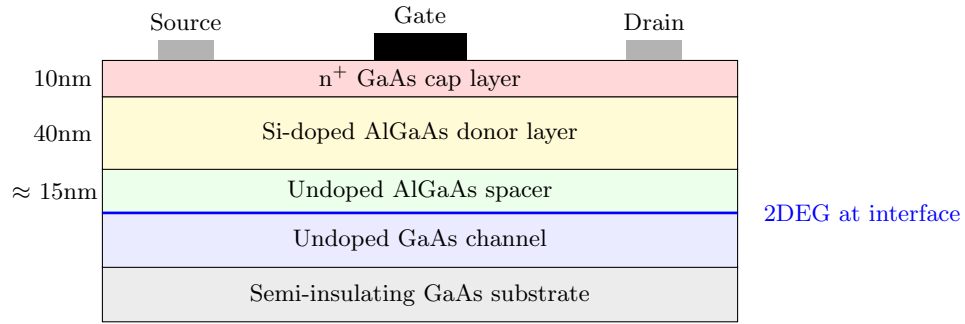


Figure 2.2: Cross-section of the GaAs/AlGaAs HEMT used in this project (not to scale).

act as electron suppliers, with carriers transferring into the GaAs channel as a result of the conduction band discontinuity at the heterointerface. This modulation-doping strategy spatially separates the ionised dopants from the conducting channel. Both the donor concentration and its thickness strongly influence the resulting sheet carrier density of the 2DEG and therefore the threshold voltage and maximum drain current of the device.

A 15 nm thick undoped AlGaAs spacer layer separates the donor layer from the GaAs channel. This spacer suppresses Coulomb scattering by physically isolating the 2DEG from ionised impurities, enabling exceptionally high electron mobility [12]. The benefit is particularly pronounced in HEMT devices operating at cryogenic temperatures, where phonon scattering is reduced and impurity scattering would otherwise dominate transport behaviour [13]. Increasing the spacer thickness improves mobility but weakens electrostatic coupling between the gate and the channel; this trade-off is a fundamental consideration in heterostructure design.

In this project, the thickness of the doped AlGaAs donor layer beneath the gate represents the primary controllable parameter through which the threshold voltage may be engineered. The following section therefore examines how gate bias modulates channel charge and determines the operational mode of the device.

2.3 Gate Control of Channel Charge and Threshold Voltage

The electrical behaviour of the GaAs/AlGaAs HEMT is governed by the field-effect interaction between the Schottky gate and the 2DEG. When a metal gate is deposited onto the semiconductor surface, a depletion region forms beneath it; the depth of this region is modulated by the applied gate voltage (V_G). This modulation controls the sheet carrier density (n_s) in the channel and, consequently, the drain current [14].

2.3.1 Charge Distribution and Effective Gate Separation

In a simple charge-control description of GaAs/AlGaAs HEMTs, the 2DEG is treated as an infinitesimally thin sheet of charge located at the GaAs/AlGaAs heterointerface, analogous to a parallel-plate capacitor; with the gate and channel acting as plates and the AlGaAs donor layer acting as a 'dielectric'. [15] This approximation is adequate for devices with relatively thick barrier layers. As the gate is recessed and the gate-to-channel separation is reduced towards the nanometre scale required for enhancement-mode operation, however, the assumption of a

sheet-like charge distribution becomes increasingly inaccurate.

In practice, electrons forming the 2DEG occupy a quantised triangular potential well on the GaAs side of the heterointerface, resulting in a charge distribution of finite thickness. The centre of this distribution is displaced from the interface, increasing the effective electrostatic separation between the gate and the channel.

Lee *et al.* account for this behaviour by introducing an effective gate-to-channel separation of $d + \Delta d$, where Δd represents a quantum-mechanical correction associated with the finite spatial extent of the electron wavefunction. For GaAs/AlGaAs HEMTs, Δd is of order 80 Å [15]. This correction may be incorporated into the transconductance parameter β , which relates changes in gate voltage to changes in drain current through the gate-controlled channel capacitance:

$$\beta = \frac{\epsilon \mu W}{(d + \Delta d)L} \quad (2.1)$$

ϵ : Permittivity of the barrier; μ : Electron mobility; W, L : Gate width and length; d : Physical gate-to-channel separation; Δd : Effective charge-centroid offset.

For the depletion-mode devices fabricated in Semester 1, the total electrostatic separation between the gate and the 2DEG is sufficiently large that this quantum correction represents a small fraction of the overall spacing. As the donor layer is thinned towards the dimensions required for enhancement-mode operation, the correction becomes a more significant contribution. Its relevance in this project lies in informing the practical recess depth required to shift the threshold voltage to positive values, and an important consideration to the transconductance of the HEMT.

2.3.2 Threshold Voltage and Mode Determination

The distinction between depletion-mode (D-mode) and enhancement-mode (E-mode) operation is determined by the balance between the built-in potential of the Schottky gate and the thickness of the doped AlGaAs donor layer. Within the analytical framework presented by Lee *et al.* [15], the threshold voltage (V_{off}) may be expressed as

$$V_{off} = \phi_b - \Delta E_c - \frac{q N_d d_d^2}{2\epsilon} \quad (2.2)$$

ϕ_b : Schottky barrier height; ΔE_c : Conduction band offset; q : Elementary charge; N_d : Donor concentration; d_d : Donor thickness; ϵ : Permittivity of the AlGaAs barrier.

This quadratic dependence of V_{off} on donor layer thickness highlights the strong sensitivity of the threshold voltage to small variations in etched depth. This relationship directly underpins the two operational modes explored in this project.

Depletion Mode (Semester 1): The baseline devices fabricated in Semester 1 employed the full epitaxial structure beneath the gate. Under these conditions, the zero-bias depletion region does not extend to the heterointerface, leaving the 2DEG populated ($V_{off} < 0$) and the device normally on, such that a negative gate voltage is required to pinch off the channel [5].

Enhancement Mode (Semester 2 Objective): To achieve normally-off behaviour ($V_{off} > 0$), the donor layer must be thinned until the depletion region fully intersects the 2DEG at zero gate bias. For a donor concentration of $1 \times 10^{18} \text{ cm}^{-3}$, the critical donor thickness is approximately 25 nm [15]; this finding by Lee *et al.* serves as a well-grounded guide for the similar GaAs/AlGaAs material structure employed in this project. Recessing the gate to this depth shifts the threshold voltage, such that a positive gate bias is required to re-populate the channel and initiate conduction [9].

Together, these relationships demonstrate that achieving enhancement-mode operation in GaAs/AlGaAs HEMTs depends critically on precise control of the donor layer thickness beneath the gate, forming the physical basis for the fabrication strategy developed in Chapter 5.

CHAPTER 3

Fabrication of Depletion-Mode HEMT Devices

3.1 Overview of the Fabrication Workflow

The depletion-mode HEMTs characterised in this report were fabricated using the standard III-V device process established within the Leeds Nanotechnology Cleanroom. This workflow follows the routine sequence used for GaAs/AlGaAs transistors in this facility: it consists of 3 main stages: mesa definition, ohmic contact formation, and gate formation. This well-established procedure was followed without modification and carried out over four full-day laboratory sessions. The devices produced through this workflow provide a necessary baseline; they allow for a reliable comparison against which the behaviour of e-mode devices with different structures and dimensions can be evaluated later in this project.

Day 1 - Mesa Formation	Cleaning	Lithography	Wet Etching	Cleaning		
Day 2 - Ohmic Contacts	Cleaning	Lithography	Metal Deposition	Metal Lift-off	Cleaning	
Day 3 - Gate Formation	Annealing	Cleaning	Lithography	Metal Deposition	Lift-off	Cleaning
Day 4 - Electrical Testing	Microscope Inspection	Defect Removal	Electrical Characterisation			

Figure 3.1: Timeline overview of the four-day cleanroom workflow used to fabricate and characterise baseline depletion-mode GaAs/AlGaAs HEMTs.

3.2 Mesa Definition and Isolation Etch

The fabrication process began with the electrical isolation of individual device regions through mesa etching. Prior to lithography, the GaAs wafer was cleaned sequentially in acetone, isopropanol (IPA), and deionised water (DI) to remove organic residues and ensure sufficient photoresist adhesion. To further enhance the bonding interface between the substrate and the resist, a Surpass primer was spun onto the wafer. This priming step is critical for wet-etching applications: it modifies the surface tension to prevent the etchant from penetrating under the resist edges which would otherwise lead to the delamination of the mask. A positive photoresist (S1803) was then spun onto the sample at 3000 rpm, soft-baked on a hot-plate at 115 °C, and subsequently patterned with exposure from the Heidelberg MLA 150 Maskless Aligner (MLA) [16].

Following exposure, the sample was immersed in a caustic Microposit developer solution for 45 seconds, followed by a thorough two-stage deionised (DI) water rinse and nitrogen blow-dry to reveal the mesa patterns. The mesa pattern was then transferred into the wafer using a

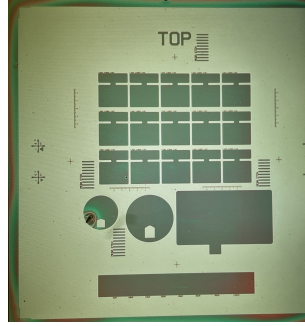


Figure 3.2: Mesa lithography mask used for defining isolated device regions (Post-Development).

wet-etch solution of sulfuric acid, hydrogen peroxide, and water ($H_2SO_4:H_2O_2:H_2O$ in a 1:8:160 ratio).

This etchant is well-suited for GaAs/AlGaAs mesa formation, as it provides a controllable, moderately fast isotropic etch that removes the active layers uniformly [17]. The etch rate is slow enough for precise timing of a shallow ≈ 200 nm recess, yet fast enough for routine device isolation.

To calibrate the process, an initial 30-second pilot etch was performed. Subsequent measurement with an Alpha-Step profilometer showed an etch depth of 114 nm, indicating an initial rate of 3.8 nm/s. To achieve the target depth of 200 nm, an additional 86 nm of material needed removal. Assuming a constant etch rate, a second etch of 22 seconds was calculated and carried out. However, the final measured depth was only 177 nm, meaning the second etch removed just 63 nm at a reduced effective rate of approximately 2.9 nm/s. Although 177 nm is adequate for device isolation, successfully penetrating the active GaAs/AlGaAs heterostructure layers, the rate discrepancy underscores the sensitivity of the wet-etch chemistry.

The main hypothesis for the reduced average etch rate in the second step is the formation of a native oxide layer on the GaAs surface during the interval when the sample was removed for profilometry. This oxide creates an “incubation period” at the start of the second etch, delaying material removal while the etchant penetrates the oxide to reach the underlying semiconductor. Since the second etch (22 seconds) was shorter than the first (30 seconds), this non-productive period accounted for a larger fraction of the total time, lowering the overall average rate.

A more specific attribution points to rapid oxidation of the AlGaAs donor layer surface during the profilometry interval. Unlike pure GaAs, which forms relatively unstable oxides, AlGaAs reacts with atmospheric oxygen to create a robust native oxide layer primarily composed of thermodynamically stable aluminium oxide (Al_2O_3), along with Ga_2O_3 and As_2O_3 [?, 18].

The $H_2SO_4:H_2O_2:H_2O$ wet-etch mechanism involves a continuous cycle: surface oxidation by the peroxide, followed by acid dissolution of the resulting oxide [17]. However, the air-grown, Al-rich passivation layer dissolves much more slowly than the transient, Al-poor oxides formed in situ during steady-state etching [18]. As a result, the second etch step includes a substantial incubation period to breach this stable Al_2O_3 cap before resuming the intrinsic etch rate of 3.8 nm/s. In a brief 22-second etch, this nonlinear delay constitutes a significant portion of the process, reducing the calculated average rate.

3.3 Ohmic Contact Formation

The formation of low-resistance Ohmic contacts is a critical prerequisite for accessing the two-dimensional electron gas (2DEG) within the heterostructure. When metal is deposited directly onto the GaAs cap, the interface naturally forms a Schottky barrier due to Fermi-level pinning at the surface states: this creates a rectifying junction that restricts carrier injection. To overcome this, a specific metallisation scheme and thermal processing step are required to facilitate field-emission (tunnelling) conduction. The objective is to create a heavily doped n^+ region beneath the contact pads that extends down towards the channel; this high doping density significantly narrows the depletion width of the barrier, allowing electrons to tunnel freely between the metal and the semiconductor with linear $I - V$ characteristics.

To define these contacts, a dual-layer photoresist stack comprising Lift-Off-Resist (LOR) beneath S1803 was employed. A single photoresist layer typically develops near-vertical sidewalls: this causes the evaporated metal to form a continuous film across both the wafer surface and the photoresist edges, making lift-off unreliable and often leaving metallic "fences" around the contact perimeter. The dual-layer stack mitigates this by exploiting the higher developer solubility of the LOR base layer. During development, the LOR recedes laterally more than the S1803 above it, producing a distinct undercut profile. This physical overhang breaks the continuity of the deposited metal film, ensuring that the metal on top of the photoresist is electrically and mechanically separated from the device contacts. Consequently, solvent-based lift-off yields sharply defined openings free of edge defects.

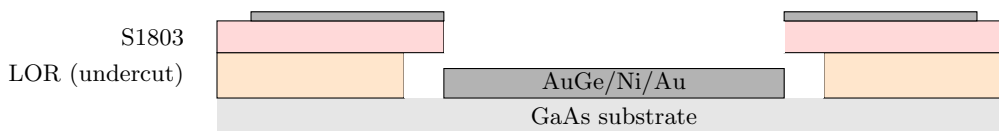


Figure 3.3: Dual-layer LOR/S1803 resist stack after development showing the wider LOR opening that creates an undercut for clean metal lift-off.

Following lithography, the Ohmic contacts were formed by thermally evaporating a AuGeNi eutectic pellet. This alloy is commonly used for n-type GaAs Ohmic contacts because each element plays a specific role during annealing. Germanium diffuses into the GaAs and acts as a donor, creating a highly doped region beneath the contact. Nickel improves the uniformity and adhesion of the metal film and supports the diffusion process, whilst gold provides a low-resistance conduction path between the contact pad and the semiconductor.

To activate the contacts, the sample underwent Rapid Thermal Annealing (RTA). It is during this thermal cycle that the transition from rectifying to Ohmic behaviour occurs. The heat drives the Germanium atoms to diffuse through the AlGaAs layers and into the GaAs lattice, locally creating the n^+ doped region. This effectively "thins" the potential barrier at the interface, establishing a low-resistance tunnelling path that connects the drain and source electrically to the 2DEG.



Figure 3.4: Ohmic contacts under x5 microscopy following lift-off.

3.4 Gate Formation and Final Inspection

With the Ohmic contacts established and annealed, a third photolithography step was performed to define the Schottky gate region. As with the previous stage, a dual-layer resist stack was utilised to ensure a clean lift-off. The gate electrode was aligned centrally between the source and drain contacts; its placement is critical, as it provides the electrostatic control required to modulate the carrier density within the channel.

Following development, a Ti/Au stack was evaporated to form the gate electrode. In this configuration, the Titanium (Ti) layer serves as a vital adhesion layer to the semiconductor surface, while the thick Gold (Au) layer provides a stable Schottky barrier and minimises gate resistance. Unlike the Ohmic contacts, the gate metallisation is not annealed; preserving the Schottky barrier height is essential for minimising leakage current and maintaining effective depletion control over the channel.

After metallisation, the resist was removed using solvent lift-off to reveal the final device structure. A small-number of lift-off defects were observed under the microscope (Figure 3.5) and subsequently removed using brief, gentle ultrasonic cleaning. The removal of these artefacts is essential for device reliability: they can otherwise cause electrical short circuits, particularly around the sensitive gate edges.

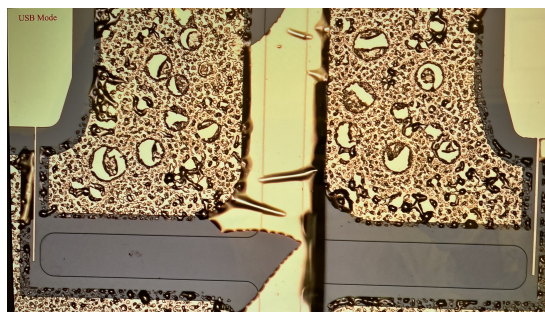


Figure 3.5: Lift-off defects (circled) observed on completed devices prior to ultrasonic cleaning.

The completed devices were examined using optical microscopy to ensure process integrity. The mesas exhibited clean, well-defined sidewalls; the Ohmic pads were properly positioned; and the gate electrodes showed sharp, uninterrupted edges with no evidence of bridging or residual flakes. Overall, the fabrication process yielded a high-quality array of depletion-mode HEMTs suitable for subsequent electrical characterisation.

3.5 Summary of Fabricated Device Set

The fabrication process described in this chapter yielded a structured array of depletion-mode GaAs/AlGaAs HEMT devices. Devices were arranged in a regular grid across the wafer, allowing multiple transistors with differing geometries to be fabricated simultaneously under identical process conditions.

In total, fifteen HEMT devices were produced and subsequently characterised. These comprised five distinct gate geometries, with three nominally identical devices fabricated for each geometry. The primary parameter varied across the grid was the effective gate length (ranging from $10\mu\text{m}$ to $4\mu\text{m}$ in Figure 3.6), whilst all other device features, including epitaxial layer structure, gate width, source-drain spacing, and metallisation scheme, were held constant.

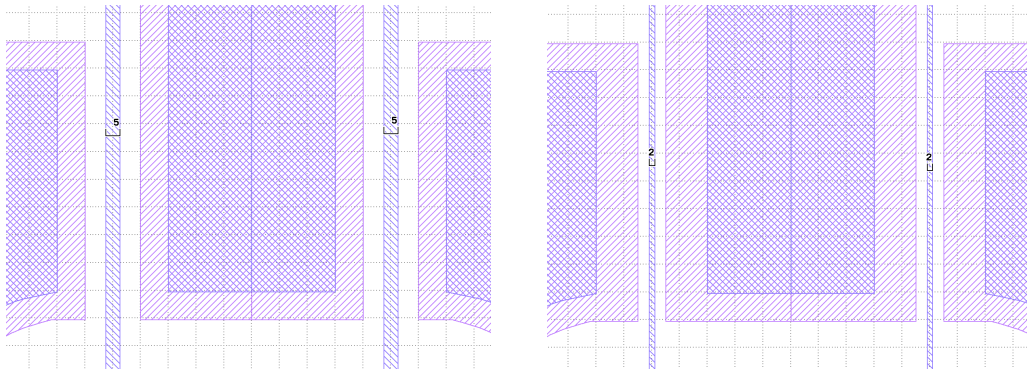


Figure 3.6: Mask layout demonstrating the extremes of the gate length range. $10\mu\text{m}$ (Left) and $4\mu\text{m}$ (Right).

The three repeated devices for each gate geometry were fabricated. This repetition provides a direct measure of device-to-device variability and fabrication repeatability, allowing both average behaviour and spread to be assessed during electrical analysis.

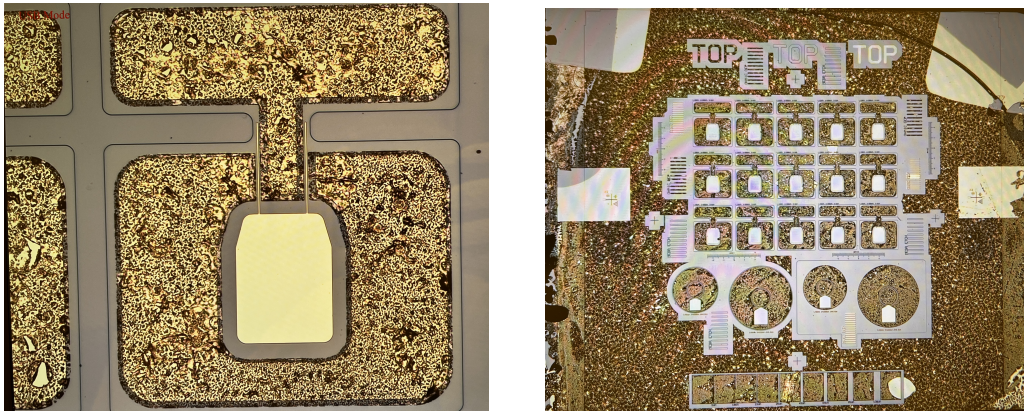


Figure 3.7: Final Fabricated HEMT Device with Gate Length $2\mu\text{m}$ (Left). Array of fabricated depletion-mode GaAs/AlGaAs HEMT devices (Right).

This device set forms a well-controlled baseline. The electrical characterisation presented in Chapter 4 serves not only to assess individual depletion-mode device performance, but also to validate the suitability of the unrecessed process as a reference for subsequent enhancement-mode device fabrication via controlled gate recession.

CHAPTER 4

Electrical Characterisation and Analysis

This chapter presents the electrical characterisation and critical analysis of the depletion-mode GaAs/AlGaAs HEMT devices fabricated using the unrecessed process described in Chapter 3. The measurements reported here are intended to establish a robust baseline against which subsequent enhancement-mode devices may be evaluated next semester.

Characterisation was performed across the full array of fifteen devices, comprising five gate lengths with three nominally identical repeats of each geometry. The analysis therefore focuses not only on individual device performance, but also on geometry-dependent trends and device-to-device variability arising from fabrication and electrostatic effects.

4.1 Measurement Setup

Electrical measurements were carried out using a Keysight B1500A semiconductor device parameter analyser [16] under ambient laboratory conditions at room temperature (21 °C).

Output characteristics were obtained by sweeping the drain–source voltage (V_{DS}) at fixed gate voltages in order to examine linear-region conduction and current saturation behaviour. Transfer characteristics were measured by sweeping the gate–source voltage (V_G) at a fixed drain bias chosen to ensure operation within the saturation regime. From these measurements, the drain current (I_D), transconductance (g_m), and threshold behaviour were extracted.

Channel resistance was determined from the low-bias linear region of the output characteristics, where the device operates in the ohmic regime. A consistent drain–source voltage range was used for all devices to enable direct comparison between gate geometries. Transconductance was calculated numerically from the measured I_D – V_G data using finite-difference methods, with identical differentiation procedures applied throughout.

For each gate geometry, electrical characteristics were averaged across three nominally identical devices to provide a representative response while suppressing random device-to-device variation.

4.2 Output Characteristics

The output characteristics of the fabricated depletion-mode HEMTs were measured to assess channel conduction, current saturation behaviour, and gate-length scaling within the unrecessed process. Figure 4.1 shows the measured I_D – V_{DS} characteristics, presented both as averaged responses for each gate geometry and as a direct comparison between the shortest and longest gate-length devices.

Across all gate geometries, the devices exhibit a clear linear region at low drain–source voltage followed by well-defined current saturation at higher V_{DS} . The linear-region behaviour confirms low-resistance ohmic contacts and uniform channel formation across the device array. The onset

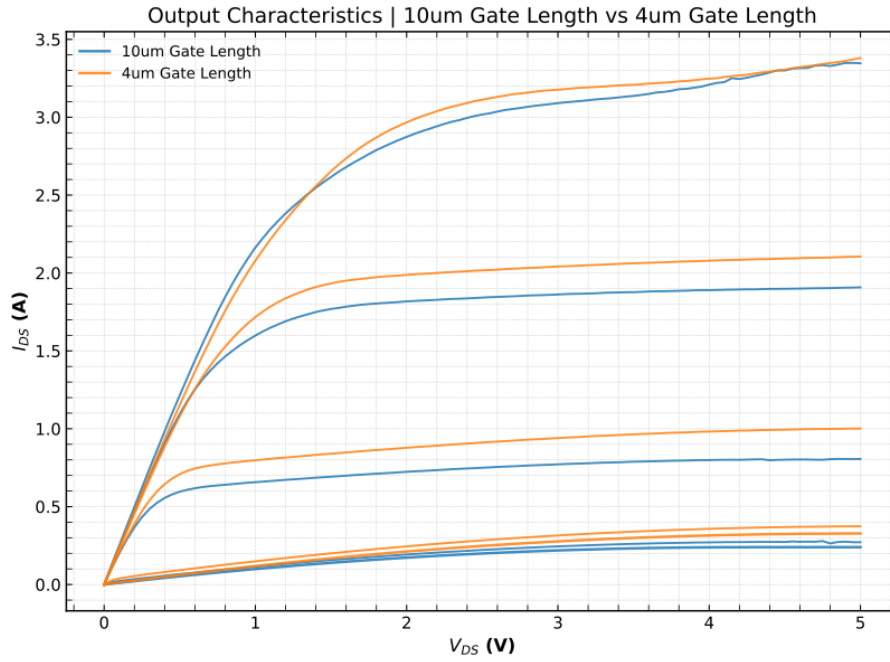


Figure 4.1: Measured output characteristics (I_D – V_{DS}) of the fabricated depletion-mode GaAs/AlGaAs HEMTs. Curves show averaged responses for each gate length, with comparison between the shortest (Orange) and longest (Blue) gate geometries highlighted.

of saturation is consistent with electrostatic pinch-off near the drain, indicating effective gate control of the channel.

A systematic dependence of drain current on gate length is observed. Devices with shorter gate lengths exhibit increased drain current at a given gate bias, consistent with reduced channel resistance and enhanced current drive. This trend is monotonic across the full set of geometries and is preserved in the averaged characteristics, indicating that it reflects intrinsic gate-length scaling rather than random device-to-device variation.

Importantly, no significant degradation in saturation behaviour is observed for the shortest gate devices within the measured bias range. The saturation regions remain relatively flat across all geometries, suggesting that short-channel effects such as drain-induced barrier lowering are not dominant for the gate lengths investigated. This supports the validity of treating these devices as long-channel references for subsequent electrostatic and threshold-voltage analysis.

Comparison between the three nominally identical devices for each gate geometry reveals a high degree of repeatability in the output characteristics. The close overlap of curves within each geometry indicates that process-induced variability, including contact resistance fluctuations and local heterostructure non-uniformity, is small relative to the systematic trends introduced by gate-length variation.

Overall, the output characteristics demonstrate predictable scaling with gate length, robust current saturation, and good repeatability across the device array. These results confirm the electrical integrity of the baseline depletion-mode fabrication process and establish a reliable reference against which the impact of gate recession and enhancement-mode operation can be assessed in later stages of this project.

4.3 Gate Transfer Characteristics

The gate transfer characteristics were measured to evaluate electrostatic control of the channel, threshold behaviour, and the influence of gate length on current modulation in the unrecessed depletion-mode devices. Figure 4.2 presents the measured I_D - V_G characteristics, shown as averaged responses for each gate geometry together with a direct comparison between the shortest and longest gate-length devices.

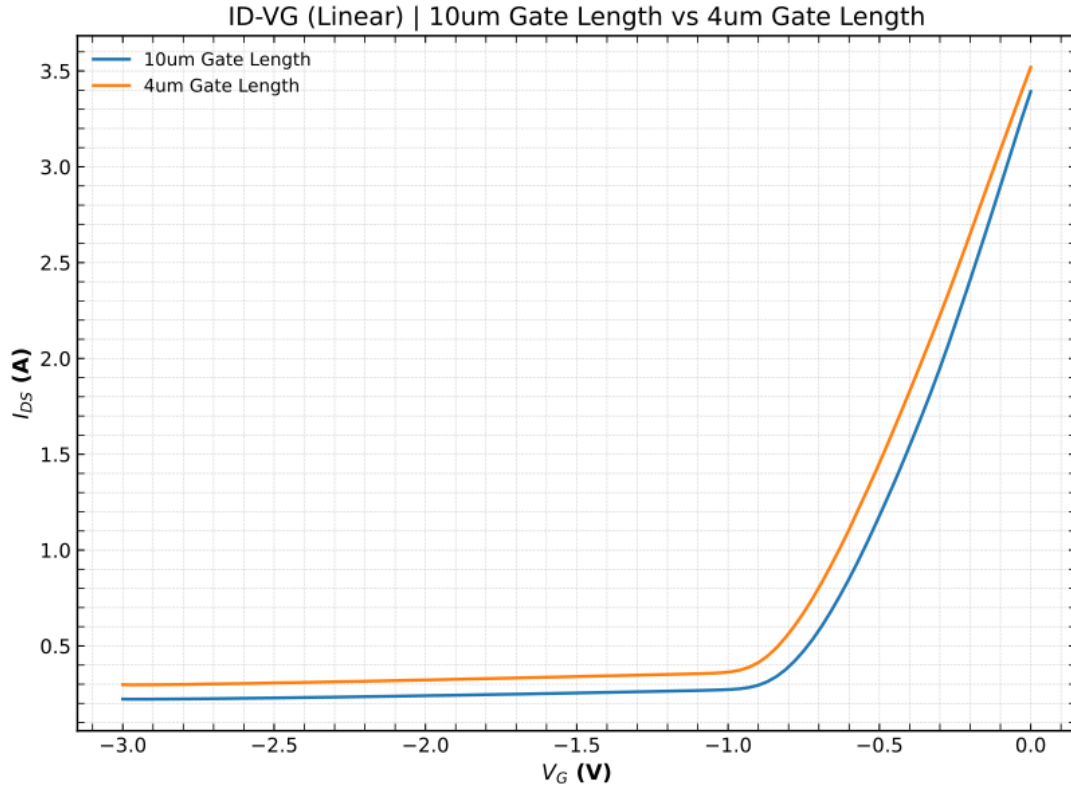


Figure 4.2: Measured transfer characteristics (I_D - V_G) of the fabricated depletion-mode GaAs/AlGaAs HEMTs. Curves show averaged responses for each gate length, with comparison between the shortest and longest gate geometries highlighted.

All devices exhibit clear depletion-mode behaviour, with substantial drain current present at zero gate bias and progressive channel pinch-off as the gate voltage is made more negative. The smooth and monotonic dependence of I_D on V_G across all geometries indicates effective gate control of the two-dimensional electron gas and an absence of significant trapping or hysteresis effects within the measured bias range.

The extracted threshold voltage is broadly consistent across gate lengths, with only modest variation observed between geometries. This behaviour is expected for long-channel, unrecessed HEMTs in which the threshold voltage is primarily set by the vertical electrostatics of the heterostructure rather than lateral device dimensions. The weak dependence of threshold voltage on gate length therefore supports the assumption that short-channel effects are minimal in the present device set.

At a given gate overdrive, devices with shorter gate lengths exhibit higher drain current, reflecting reduced channel resistance and increased current drive. This trend is consistent with the scaling observed in the output characteristics and confirms that lateral gate-length varia-

tion primarily modulates channel conductance rather than the onset of channel formation. The similarity of the subthreshold-to-on transition across geometries further indicates that the electrostatic coupling between the gate and the channel is not significantly degraded for shorter gate lengths within the range investigated.

Repeatability across the three nominally identical devices for each geometry is high, with close overlap observed in the averaged transfer characteristics. The limited spread in both threshold voltage and slope suggests that fabrication-induced variability, such as fluctuations in donor density or Schottky barrier quality, is small relative to the systematic effects of gate length scaling. This level of uniformity is particularly important in the context of subsequent gate recessing, where small variations in vertical geometry are expected to produce comparatively large shifts in threshold voltage.

Overall, the transfer characteristics confirm strong and consistent gate control of the channel, stable depletion-mode operation, and weak gate-length dependence of threshold voltage. These results validate the use of the present device set as a reliable electrical baseline for assessing the impact of controlled gate recession in enhancement-mode devices fabricated in the second phase of this project.

4.4 Gate Leakage Behaviour

Gate leakage measurements were performed to assess the integrity of the Schottky gate contact and to identify any dependence of leakage behaviour on gate length. Figure 4.3 presents the measured gate current as a function of drain–source voltage, shown as averaged responses for each gate geometry together with a direct comparison between the shortest and longest gate-length devices.

Across all devices, the measured gate currents remain low over the full range of applied drain bias, indicating the formation of a high-quality Schottky barrier and good isolation between the gate electrode and the channel. The absence of abrupt increases in gate current suggests that gate metallisation and lift-off were successful, with no evidence of widespread gate shorts or catastrophic edge defects.

A weak dependence of gate leakage on gate length is observed, with shorter gate devices exhibiting slightly higher leakage currents. This trend is consistent with increased electric-field crowding at the gate edges as the lateral dimensions are reduced, rather than a degradation of the Schottky barrier itself. Importantly, the magnitude of the leakage remains sufficiently small that it does not measurably influence the transfer or output characteristics discussed in the preceding sections.

Repeatability across the three nominally identical devices for each geometry is high, with minimal spread in leakage behaviour observed within each gate length. This consistency indicates that gate leakage is governed primarily by systematic geometric effects rather than random process-induced defects or localised surface damage.

From a process perspective, the low and repeatable gate leakage observed in the unrecessed devices provides confidence in the quality of the gate formation step. However, the modest increase in leakage for shorter gates highlights a potential sensitivity to electric-field enhancement

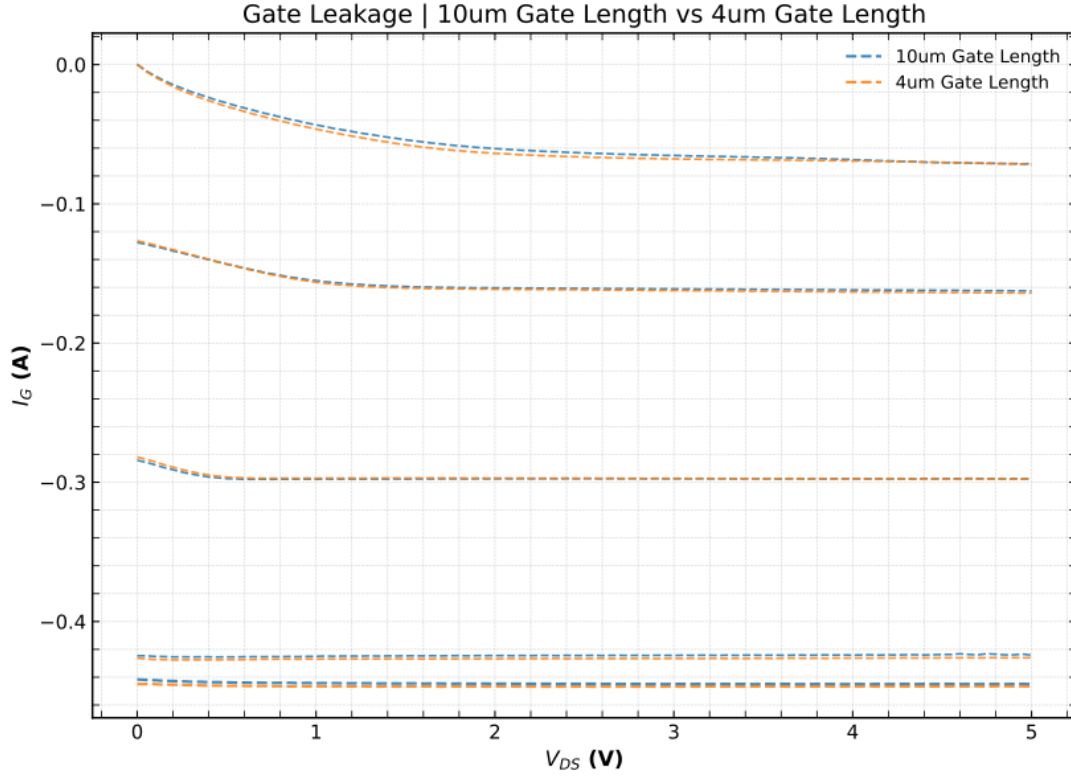


Figure 4.3: Measured gate leakage characteristics of the fabricated depletion-mode GaAs/AlGaAs HEMTs. Curves show averaged responses for each gate length, with comparison between the shortest and longest gate geometries highlighted.

that will become more pronounced as the gate-to-channel separation is reduced during gate recession. As such, gate leakage represents a key constraint that must be monitored carefully in the development of enhancement-mode devices in the second phase of this project.

4.5 Threshold-Voltage Extraction

4.6 Variability, Non-Idealities and Interpretation

4.7 Implications for Process Control and Device Behaviour

CHAPTER 5

Exploration of Enhancement-Mode HEMT Devices

- 5.1 Motivation for Enhancement-Mode Operation
- 5.2 Mechanisms for Achieving Enhancement Mode
- 5.3 Challenges of Gate Recessing Without an Etch-Stop Layer
- 5.4 Proposed Fabrication Strategy for Semester Two
- 5.5 Ideal Outcomes and Target Behaviour
- 5.6 Logic Motivation and Stretch Goal

CHAPTER 6

Reflection and Project Management

6.1 Summary of Progress

6.2 Challenges and Adaptations

6.3 Skills Developed

6.4 Lessons for Semester Two

CHAPTER 7

Semester Two Plan

7.1 Gantt Chart

7.2 Planned Technical Deliverables

7.3 Risk Mitigation Strategy

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APPENDIX A

Additional Figures

APPENDIX B

Code and Process Flow